AIVOA.AI — Pharmaceutical Root Cause Analysis

System Specification & Architecture Overview Document

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System Type: AI-Assisted RCA & CAPA	**Regulatory Scope:** 21 CFR Part 11, ICH Q7/Q9(R1)/Q10	**Status:** Approved & Verified

1. Executive Summary & Problem Statement

In pharmaceutical Active Pharmaceutical Ingredient (API) and drug product manufacturing, quality deviations—such as Out-of-Specification (OOS) analytical assay results, bioburden excursions, crystallization yield failures, or critical equipment breakdowns—trigger formal non-conformance investigations mandated by global regulatory agencies (US FDA 21 CFR Part 211/Part 11, EU cGMP Volume 4, and ICH Guidelines Q7, Q9(R1), and Q10).

Historically, pharmaceutical investigations suffer from significant systemic issues:

- **Severe Latency & Transcription Overhead:** Investigators spend between 4 to 8 hours transcribing physical batch records, shift notes, and email chains into disjointed QMS/ERP systems before initiating causal analysis.
- **Cognitive Bias & Superficial Root Causes:** Investigations frequently default to generic 'Operator Error' or 'Retesting within specification' rather than uncovering underlying systemic or mechanical failure modes.
- **Disconnected CAPAs:** Corrective actions often lack causal linkage to specific evidence items or root causes, treating immediate cleanup/containment as permanent preventive measures.
- **Audit Deficiencies:** Incomplete investigation rationales, missing evidence documentation, and unverified causal hypotheses represent major categories of FDA Form 483 inspection observations.

Platform Mission: AIVOA.AI was engineered as a domain-grounded Root Cause Analysis (RCA) and CAPA platform bridging multi-modal intake with formal Quality Risk Management. It functions as an advisory co-pilot that structures raw evidence, constructs standard causal hypotheses (5 Whys and 6M Fishbone), flags cGMP evidence gaps, and drafts targeted CAPA proposals while strictly preserving a human-in-the-loop regulatory audit trail.

2. Project Objectives

- **1. Automate Multi-Source Deviation Ingestion:** Eliminate manual re-keying by parsing unstructured inputs across three distinct modalities: official deviation PDF reports, unstructured laboratory/shift notes, and escalation email notifications.
- **2. Standardize Pharmaceutical RCA Frameworks:** Enforce rigorous Root Cause Analysis methodologies compliant with ICH Q9(R1), specifically supporting 5 Whys sequential causal chains, Ishikawa 6M fishbone categorization (Man, Machine, Material, Method, Measurement, Milieu), and combined hybrid analysis.
- **3. Automate cGMP Evidence Gap Identification:** Proactively analyze incident narratives to identify missing batch manufacturing records (BMR), equipment calibration certificates, sensor historians, and raw material Certificates of Analysis (CoA) before regulatory closure.

- **4. Differentiate Corrective vs. Preventive Actions (ICH Q10):** Ensure strict demarcation between short-term corrective containment actions (lot isolation, re-calibration) and systemic preventive controls (engineering interlocks, SOP revisions).
- **5. Enforce Human Oversight Safeguards:** Maintain an immutable boundary where AI outputs remain advisory proposals. Only authenticated human quality investigators can confirm root causes, modify investigation findings, approve CAPAs, and execute final sign-off.
- **6. Ensure Regulatory Traceability & Persistence:** Persist all investigation records, evidence items, causal chains, and review histories in a relational PostgreSQL database with full referential integrity and audit trail tracking.

3. Target Users & Personas

Role / Persona	Responsibilities in System	Primary Workflows Utilized
Quality Assurance (QA) Specialist	Reviews deviation notifications, logs initial incidents, tracks investigation progression, and performs formal regulatory review and sign-off.	Multi-modal Intake, Investigation Workspace, Step 4 Formal Review & Sign-Off, Saved Investigations Ledger.
Manufacturing Investigation Engineer	Analyzes process excursions, verifies physical equipment anomalies, tests causal hypotheses, and gathers missing batch records.	Step 1 Deviation Parameters, Step 2 Root Cause Analysis (5 Whys / 6M), Evidence Gap Tracker.
Quality Control (QC) Analytical Lead	Investigates laboratory Out-of-Specification (OOS) results, chromatographic anomalies, and retesting protocols.	Step 1 Critical Quality Attributes (observed vs. expected specs), Evidence Gap Tracker.
Regulatory & Compliance Auditor	Audits investigation packages during regulatory inspections (FDA, EMA) or internal quality audits.	Audit Trail Viewer, Saved Investigations Historical Detail, Legal & GxP Disclosures.

4. Project Scope & Operational Boundaries

In-Scope Capabilities:

- **Multi-Modal Document Parsing:** Native PDF text extraction (PyMuPDF/pdfplumber) with magic byte validation, plain text intake, and email message parsing.
- **AI-Powered Structured Extraction:** LangGraph orchestration with Groq Llama-3.3-70B (temperature 0.1) extracting Critical Quality Attributes (batch ID, product name, equipment ID, process stage, specs, containment).
- **Domain-Grounded Fallback Engine:** Deterministic regex-based extraction pipeline for zero-downtime offline execution.
- **Interactive Investigation Workspace:** 4-step wizard interface allowing in-place editing, visual .field-ai-highlight cues, and dynamic methodology switching.
- **RCA Methodologies:** Autonomous generation and interactive editing of 5 Whys logical chains and 6M Fishbone factors.
- **CAPA Management:** Automatic generation of risk-grounded Corrective and Preventive actions with owner assignment, timelines, and effectiveness checks.
- **Relational Persistence:** Full relational storage across 8 tables in PostgreSQL / Supabase using SQLAlchemy ORM with atomic transactions.
- **Audit Trails:** Recording of investigation lifecycle state transitions (SAVED, HUMAN_REVIEW, UPDATED, EVIDENCE_STATUS_UPDATED).

Out-of-Scope / Operational Boundaries (Current Release):

- **Direct SCADA / Historian Ingestion:** Live polling from OSIsoft PI / DeltaV historians is not automated; data must be ingested as PDF, logs, or text.
- **Enterprise Single Sign-On (SSO):** OAuth2/OIDC/SAML2 authentication is not implemented; system operates with single-tenant role-tagged sessions.
- **Cryptographic Dual-Key PKI Signatures:** 21 CFR Part 11 Subpart C certified digital signature certificates are represented by documented review workflows rather than smart-card/PKI hashes.
- **Direct Third-Party QMS Auto-Push:** Webhook synchronization into external platforms (Veeva Vault, TrackWise) requires manual export or custom API connector.

5. Core Feature Matrix & Verification Evidence

Feature Capability	Description	Status	Verification Evidence
PDF Upload Intake	Multipart file upload with magic byte validation (%PDF-) and 10MB limit.	**Implemented & Verified**	Verified via test_pdf_upload_endpoint_valid in pytest suite and browser audit.
Text & Notes Intake	Ingestion of raw pasted text or observation logs via JSON payload.	**Implemented & Verified**	Verified via test_text_intake_success and test_text_intake_json_success.
Email Report Intake	Subject and body ingestion parsing sender, incident notifications, and attachments.	**Implemented & Verified**	Verified via test_email_intake_success and test_email_intake_json_success.
Groq Llama-3.3 Extraction	Extraction of 14 structured pharmaceutical parameters via prompt engineering.	**Implemented & Verified**	Verified via compiled_rca_graph and automated test suite.

Feature Capability	Description	Status	Verification Evidence
Deterministic Fallback	Regular expression and keyword parser ensuring zero-downtime test and offline capability.	**Implemented & Verified**	Verified via _fallback_extract_incident across all unit test passes.
Interactive Form Auto-Fill	Dynamic pre-population of Step 1 form fields with visual .field-ai-badge indicators.	**Implemented & Verified**	Verified in browser UI audit session with Redux state updates.
5 Whys RCA Analysis	Sequential 5-tier question-answer causal drill-down.	**Implemented & Verified**	Verified via /api/v1/rca/preview with method='5_whys'.
6M Fishbone Analysis	Categorization into Man, Machine, Material, Method, Measurement, Milieu.	**Implemented & Verified**	Verified via /api/v1/rca/preview with method='6m_fishbone'.
Combined RCA (Both)	Simultaneous generation of both 5 Whys and 6M Fishbone models.	**Implemented & Verified**	Verified via test_rca_preview_generation_both.
Evidence Gap Tracking	Automated identification of missing cGMP records (BMRs, calibration logs).	**Implemented & Verified**	Verified in EvidenceItem models and LangGraph node output.
CAPA Recommendations	Distinct generation of Corrective Actions vs. Preventive Actions with effectiveness checks.	**Implemented & Verified**	Verified via test_capa_preview_generation.
Human Review & Sign-off	Human confirmation safeguards (is_human_confirmed, review notes, reviewer name).	**Implemented & Verified**	Verified via test_investigation_lifecycle_and_safeguards.
PostgreSQL Persistence	Atomic transaction saving investigation parent and all 7 related child tables.	**Implemented & Verified**	Verified via POST /api/v1/investigations with cascade relationships.
Search & Filtering	Filtering saved investigations by status and full-text keyword search.	**Implemented & Verified**	Verified via GET /api/v1/investigations?search=...